Adaptive Computation and Machine Learning

COMS 4030A and COMS7047A

PROJECT

The project for ACML requires you to analyse a dataset using machine learning techniques and write a report describing your methods and results.
This is a **group** project. You should organise yourselves into groups of size 4 or less.

1) Source a dataset of sufficient complexity and interest. See, e.g., kaggle.com or UCI machine learning laboratory, but you are free to obtain datasets from anywhere, as long as no ethics clearance is required. State clearly where the dataset was obtained.

2) Describe the contents of the dataset and your objectives with respect to the dataset.
(Do not do too much data exploration ...)

3) Describe the required preprocessing of the dataset and the split of data into train/validate/test.

4) Describe the choice of models and machine learning algorithms used. Describe the implementation of the model. Python libraries such as Keras and Pytorch are allowed. The model should be sufficiently complex. CNN's, RNN's, autoencoders, GANS, or other complex architectures may be used. Combinations of models or techniques are allowed as well.

5) Describe any experiments that were done in fine-tuning the models or the hyperparameters, or the training methods.

6) Present various graphs that were generated during the training of the model.

7) Present the results of the trained models on the test dataset, using both text and visuals, and provide a short analysis of the results.

8) Put all the relevant information from the above into a **report**.
Be sure to add the names and student numbers of all team members to the front page of the report. Only one member of the team needs to upload the report to moodle.

Important Notes:

It is good to be explorative – try a variety of models, or a number of variations of a particular model, as wells as variations of hyperparameters and regularisers. You can report on what did not work as well as what did work.

Please **do not** just use an existing model from the internet.

It is in your benefit to make it clear what your team has created and what comes directly from internet sources.

Required submission: Your report and your code.

Due date for submission: Sunday 24 May, at 23h00.